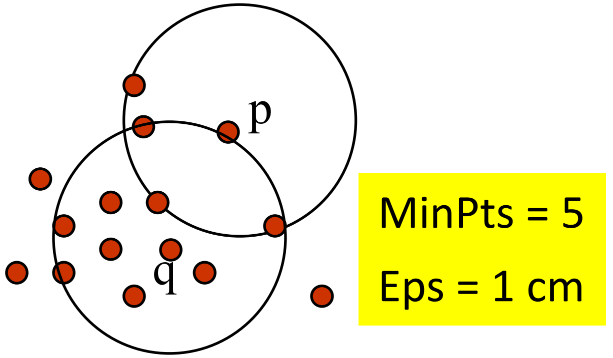
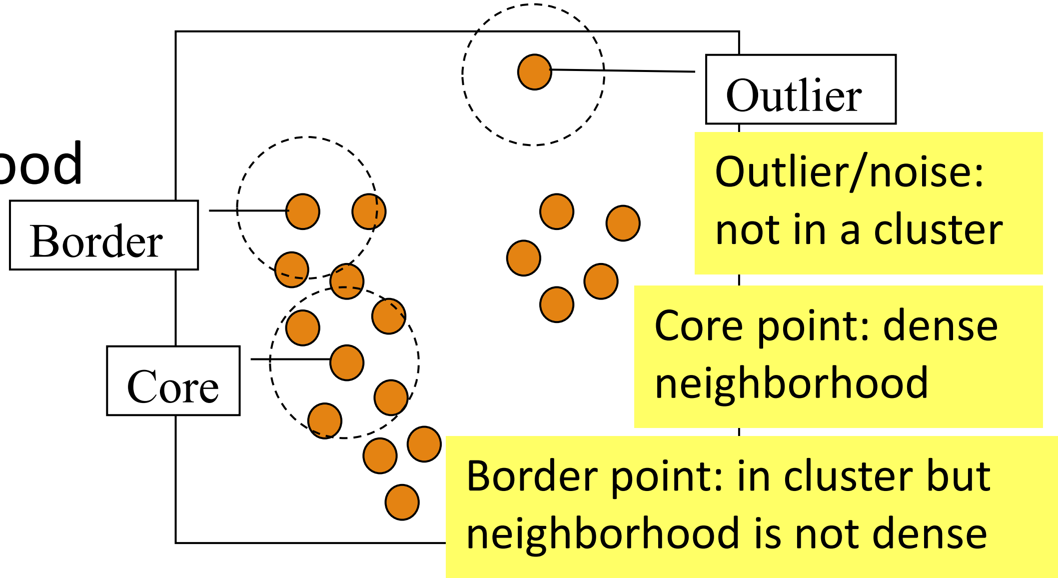
Density-Based Clustering Methods

- Clustering based on density (a local cluster criterion), such as density-connected points
- Major features:
  - Discover clusters of arbitrary shape
  - Handle noise
  - One scan (only examine the local region to justify density)
  - Need density parameters as termination condition
- Several interesting studies:
  - DBSCAN: Ester, et al. (KDD'96)
  - OPTICS: Ankerst, et al (SIGMOD'99)
  - DENCLUE: Hinneburg & D. Keim (KDD'98)
  - CLIQUE: Agrawal, et al. (SIGMOD'98) (also, grid-based)

DBSCAN: A Density-Based Spatial Clustering Algorithm

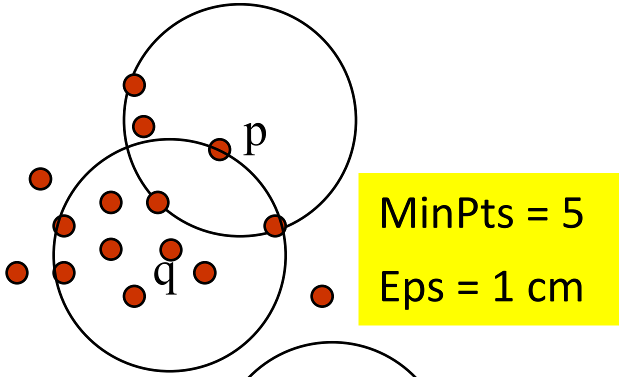
- DBSCAN (M. Ester, H.-P. Kriegel, J. Sander, and X. Xu, KDD'96)
  - Discovers clusters of arbitrary shape: Density-Based Spatial Clustering of Applications with Noise
- A density-based notion of cluster
  - A cluster is defined as a maximal set of density-connected points
  - Two parameters:
  - Eps (ε): Maximum radius of the neighborhood
  - MinPts: Minimum number of points in the
    - Eps-neighborhood of a point
- The Eps(ε)-neighborhood of a point q:
  - $N_{Eps}(q)$: {p belongs to D | dist(p, q) ≤ Eps}



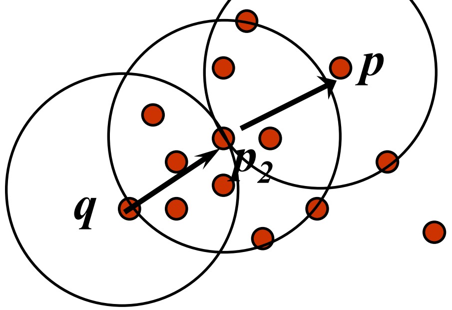


DBSCAN: Density-Reachable and Density-Connected

- Directly density-reachable:
  - A point p is directly density-reachable from a point q w.r.t. Eps (ε), MinPts if
    - p belongs to $N_{Eps}(q)$
    - core point condition: $|N_{Eps}(q)| \geq$ MinPts



- Density-reachable:
  - A point p is density-reachable from a point q w.r.t. Eps, MinPts if there is a chain of points $p_1, \ldots, p_n$, $p_1 = q$, $p_n = p$ such that p_{i+1} is directly density-reachable from p_i



- Density-connected:
  - A point p is density-connected to a point q w.r.t. Eps, MinPts if there is a point o such that both p and q are density-reachable from o w.r.t. Eps and MinPts